

Probability, Statistics and Simulation (Mid-Term)

Key: pdf, pmf, cdf ... 写取值范围!
写 0, otherwise

Lec 1 Introduction

1. Bar Charts: 柱状图/条形图
 2. Time Series Plots: 时间序列图
 3. Histogram: 直方图
 4. ①: 类别比较 (柱状图)
 5. ②: 随时间变化趋势 (一般折线)
 6. ③: 连续数据分布 (柱状图)
2. ① mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- ② median: M
- ③ mode: M_o
- ④ standard deviation: $SD = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} = \sqrt{Var(X)}$

证明: 若 $P(A) = P(A|B)$, then $P(A) = P(A|B^c)$

$P(A|B) = \frac{P(A \cap B)}{P(B)}$, $P(A) = P(A|B)$

$P(A \cap B) = P(A|B)P(B)$

$P(A) = P(A \cap B) + P(A \cap B^c) = P(A|B)P(B) + P(A|B^c)P(B^c)$

$\Rightarrow P(A \cap B^c) = P(A)(1 - P(B)) = P(A)P(B^c)$

$\Rightarrow P(A|B^c) = \frac{P(A \cap B^c)}{P(B^c)} = P(A)$

全概率: $B_1, B_2, \dots, B_n$: a partition of Ω

$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$

特别地: $P(A) = P(A \cap \Omega) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$

eg: A die is rolled repeatedly to yield "1", and then a coin is tossed the same number of times. What is the probability that heads will not appear?

A: roll 1: 得到 "1" $P(A_1) = (\frac{1}{6})^1 \cdot \frac{1}{2}$

B: heads will not appear $P(B_1) = (\frac{1}{2})^1$

$P = \sum_{i=1}^{\infty} (\frac{1}{6})^i \cdot (\frac{1}{2})^i = \frac{1}{2} \sum_{i=1}^{\infty} (\frac{1}{6})^i = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1 - (\frac{1}{6})^{n+1}}{1 - \frac{1}{6}} = \frac{1}{2} \cdot \frac{1}{5/6} = \frac{3}{5}$

等比求和: $a \cdot \frac{1 - q^n}{1 - q}$ q : 公比

注意: a 是 $i=1$ 时的值 a : 首项

否则是 a 之外

的乘积而已

Lec 6 E(X), Var(X), Cov(X, Y)

1. $E[f(x)] = \sum_i f(x_i)P(X=x_i) = \sum_i f(x_i)p(x_i)$
2. $E[ax+b] = aE[X] + b$
3. $Var(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$
4. $Var(aX+b) = a^2 Var(X)$
5. $Cov(X, Y) = E[(X-\mu_X)(Y-\mu_Y)]$
6. $Cov(X, X) = Var(X)$
7. $Cov(aX+b, Y) = aCov(X, Y)$
8. $Cov(X+Z, Y) = Cov(X, Y) + Cov(Z, Y)$
9. $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X, Y)$
10. $Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$
11. Correlation
12. If X and Y are independent:

Lec 7 Conditional pmf and cdf (given Y=y)

1. 离散: conditional pmf and cdf (given Y=y)
2. 连续: conditional pdf and cdf (given Y=y)
3. $E[X], Var(X), Cov(X, Y)$
4. $E[X] = \int_{-\infty}^{\infty} x f(x) dx$
5. $E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$
6. $E[X+Y] = E[X] + E[Y]$ (in general)
7. 其它同左 (Lec 6)

Lec 2 Foundations

1. Sample Space: Set of all possible outcomes
 2. Event: A set of outcomes in the SS
 3. Probability: Assignments of the likelihoods on these possible outcomes
- $A_n^k = \frac{n!}{(n-k)!k!}$
- $C_n^k = \frac{n!}{k!(n-k)!} = \binom{n}{k}$
- eg: 4 econ and 6 math student 两两配对
- 求: 没有 econ student 和 math student 配对成功的概率
- 总情况: $C_{10}^2 \times C_8^2 \times C_6^2 \times C_4^2 \times C_2^2$
- 成功情况: $C_4^2 \times C_6^2 \times C_2^2$
- 注意: 手算分组, 分组顺序不相类
- 小 C (2人一组) 外面有大 C (5组)

eg: A die is rolled repeatedly to yield "1", and then a coin is tossed the same number of times. What is the probability that heads will not appear?

A: roll 1: 得到 "1" $P(A_1) = (\frac{1}{6})^1 \cdot \frac{1}{2}$

B: heads will not appear $P(B_1) = (\frac{1}{2})^1$

$P = \sum_{i=1}^{\infty} (\frac{1}{6})^i \cdot (\frac{1}{2})^i = \frac{1}{2} \sum_{i=1}^{\infty} (\frac{1}{6})^i = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1 - (\frac{1}{6})^{n+1}}{1 - \frac{1}{6}} = \frac{1}{2} \cdot \frac{1}{5/6} = \frac{3}{5}$

等比求和: $a \cdot \frac{1 - q^n}{1 - q}$ q : 公比

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Lec 7 Continuous RV

1. probability density function (pdf): nonnegative function $f(x)$
2. $\int_{-\infty}^{\infty} f(x) dx = 1$
3. cumulative distribution function (cdf): $F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$
4. $f(x)$ is the derivative of $F(x) = P(X \leq x)$
5. Since F is differentiable, it's a continuous function. It has "no jumps" like the cdf of discrete random variables.
6. $f(x=x)$ is NOT $P(X=x)$ ($=0$)
7. $P(a < X < b) = P(X < b) - P(X \leq a) = \int_a^b f(x) dx$
8. eg: A chopstick of unit length is broken at random into two pieces.
9. let R = shorter piece, find pdf of R .
10. let u = the point from 0, at which the chopstick is broken.
11. $R(u) = \min\{\frac{u}{1-u}, \frac{1-u}{u}\}$
12. The event $\{u: R(u) \leq r\}$ means $\{\frac{u}{1-u} \leq r\} \cup \{\frac{1-u}{u} \leq r\}$
13. $\Leftrightarrow \{0 < u \leq \frac{r}{1+r}\} \cup \{\frac{1}{1+r} \leq u < 1\}$
14. $P(R \leq r) = P(0 < u \leq \frac{r}{1+r}) + P(\frac{1}{1+r} \leq u < 1) = \frac{2r}{1+r}$
15. Derivative of $\frac{2r}{1+r}$ is $\frac{2}{(1+r)^2}$
16. $\therefore f(x) = \begin{cases} \frac{2}{(1+x)^2} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
17. 注意: cdf $\rightarrow$ pdf
18. joint pdf $f(x, y)$: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
19. $P(x \leq a, y \leq b) = \int_{-\infty}^a \int_{-\infty}^b f(x, y) dx dy$
20. $f(x, y) = \frac{\partial^2}{\partial x \partial y} P(X \leq x, Y \leq y)$
21. marginal pdf: $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$
22. $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$
23. $P(X \leq x) = \int_{-\infty}^x f_X(x) dx$
24. $P(Y \leq y) = \int_{-\infty}^y f_Y(y) dy$
25. X, Y are independent $\Leftrightarrow P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$
26. $f(x, y) = f_X(x)f_Y(y)$
27. wait for two buses the arrival times X, Y has a joint pdf: $f(x, y) = 6e^{-(2x+y)}$ for $x > 0$ and $y > 0$
28. (1) X, Y 独立? (2) 求 pdf of $\min\{X, Y\}$
29. $f_X(x) = 6e^{-2x} \int_0^{\infty} e^{-y} dy = 2e^{-2x}$
30. $f_Y(y) = 6e^{-y} \int_0^{\infty} e^{-2x} dx = 3e^{-y}$
31. (3) let $Z = \min\{X, Y\}$
32. $P(Z \leq t) = P(X \leq t \cup Y \leq t) = \int_0^t \int_0^{\infty} f(x, y) dx dy + \int_t^{\infty} \int_0^t f(x, y) dx dy$
33. $= \int_0^t \int_0^{\infty} 6e^{-(2x+y)} dx dy + \int_t^{\infty} \int_0^t 6e^{-(2x+y)} dx dy$
34. $= \int_0^t 3e^{-y} dy + \int_t^{\infty} 3e^{-y} dy = 1 - e^{-t}$
35. $\therefore f_Z(t) = e^{-t}$ for $t > 0$
36. $\therefore P(Z \leq t) = 1 - e^{-t}$
37. $f_Z(t) = e^{-t}$ for $t > 0$

Lec 8 Special RV

1. 离散型
2. 均匀分布 $X \sim \text{Unif}(a, b)$ Uniform
3. $P(X=x) = \frac{1}{b-a+1}$ $x=a, a+1, \dots, b$
4. $E[X] = \frac{a+b}{2}$
5. $Var(X) = \frac{(b-a+1)^2 - 1}{12}$
6. 两点分布 $X \sim \text{Ber}(p)$ Bernoulli
7. $P(X=1) = p$ $P(X=0) = 1-p$
8. $E[X] = p$
9. $Var(X) = p(1-p)$
10. 二项分布 $X \sim \text{Bin}(n, p)$ Binomial
11. $P(X=k) = C_n^k p^k (1-p)^{n-k}$ $k=0, 1, \dots, n$
12. $E[X] = np$
13. $Var(X) = np(1-p)$
14. 若 $X_1, X_2, \dots, X_n$ 相互独立, 且 $X_i \sim \text{Ber}(p)$
15. 则 $S = X_1 + X_2 + \dots + X_n \sim \text{Bin}(n, p)$
16. 若 X, Y 相互独立, 且 $X \sim \text{Bin}(n, p), Y \sim \text{Bin}(m, p)$
17. 则 $X+Y \sim \text{Bin}(n+m, p)$
18. 几何分布 $X \sim \text{Geo}(p)$ Geometric
19. $P(X=k) = (1-p)^{k-1} p$ $k=1, 2, \dots$ "first success"
20. $E[X] = \frac{1}{p}$ $Var(X) = \frac{1-p}{p^2}$
21. 无记忆性: $P(X=k+1|X \geq k) = P(X=1)$
22. 泊松分布 $X \sim \text{Poi}(\lambda)$ Poisson ($\lambda > 0$)
23. $P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$ $k=0, 1, 2, \dots$
24. $E[X] = Var(X) = \lambda$
25. $\{n! \cdot P(X=n) \mid n \geq 1\}$ 则: $\text{Bin}(n, p) \rightarrow \text{Poi}(\lambda)$ ($n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$)
26. 超几何分布 $X \sim \text{HGeo}(N, K, n)$
27. 选回不放回抽样 (without replacement)
28. 两类属性互斥 ($K \leq N-K$)
29. 关注样本中某一类属性出现次数
30. $P(X=k) = \frac{C_K^k C_{N-K}^{n-k}}{C_N^n}$
31. 当 N 相对于 n 很大 $\Rightarrow$ "不放回" $\rightarrow$ "放回"
32. 则: $\text{HGeo}(N, K, n) \Rightarrow \text{Bin}(n, p)$ 其中 $p = \frac{K}{N}$
33. (二) 连续型
34. 均匀分布 $X \sim \text{Unif}(a, b)$ Uniform
35. $f(x) = \frac{1}{b-a} \mathbb{I}_{(a,b)}(x)$
36. $F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$
37. $E[X] = \frac{a+b}{2}$ $Var(X) = \frac{(b-a)^2}{12}$
38. 指数分布 $X \sim \text{Exp}(\lambda)$ Exponential ($\lambda > 0$)
39. $f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$
40. $F(x) = 1 - e^{-\lambda x}$ $x > 0$
41. $E[X] = \frac{1}{\lambda}$ $Var(X) = \frac{1}{\lambda^2}$
42. Property 1: memoryless 无记忆性
43. $P(X > s+t | X > s) = P(X > t)$
44. $P(X > s+t) = \int_s^{\infty} \int_t^{\infty} \lambda e^{-\lambda(x+y)} dx dy = e^{-\lambda(s+t)}$
45. $P(X > s) = \int_s^{\infty} \lambda e^{-\lambda x} dx = e^{-\lambda s}$
46. Property 2: $X_1 \sim \text{Exp}(\lambda_1), \dots, X_n \sim \text{Exp}(\lambda_n)$ 相互独立
47. let $Y = \min\{X_1, \dots, X_n\}$, then $Y \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$

Lec 3 Conditional Probability

1. $P(A|B) = \frac{P(A \cap B)}{P(B)}$
2. $P(A \cap B) = P(A|B)P(B)$
3. $P(A \cap B \cap C) = P(A|B \cap C)P(B \cap C)$
4. multiplication rule
5. eg: 转轮盘游戏 $N \Rightarrow C_1 \Rightarrow C_2 \Rightarrow S$
6. 已知 S , 求在 C_k 的概率:
7. $P(A|B) = \frac{P(A \cap B)}{P(B)}$
8. 注意: 隐含条件 (没在另一块去)
9. 用条件概率证明 Independence
10. A, B independent $\Leftrightarrow P(A \cap B) = P(A)P(B)$
11. $\Leftrightarrow P(A) = P(A|B)$
12. $\Leftrightarrow P(B) = P(B|A)$

eg: 证明: 若 $P(A) = P(A|B)$, then $P(A) = P(A|B^c)$

$P(A|B) = \frac{P(A \cap B)}{P(B)}$, $P(A) = P(A|B)$

$P(A \cap B) = P(A|B)P(B)$

$P(A) = P(A \cap B) + P(A \cap B^c) = P(A|B)P(B) + P(A|B^c)P(B^c)$

$\Rightarrow P(A \cap B^c) = P(A)(1 - P(B)) = P(A)P(B^c)$

$\Rightarrow P(A|B^c) = \frac{P(A \cap B^c)}{P(B^c)} = P(A)$

全概率: $B_1, B_2, \dots, B_n$: a partition of Ω

$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$

特别地: $P(A) = P(A \cap \Omega) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$

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Lec 4 Bayes' Formula

1. 应用: hypothesis 可证可错
2. defendant is guilty? someone has disease?
3. $H = \{\text{true}\}$ $H^c = \{\text{false}\}$
4. 公式: $P(H|E) = \frac{P(E|H)P(H)}{P(E)}$
5. $P(E) = P(E|H)P(H) + P(E|H^c)P(H^c)$
6. 计算 $P(H|E)$ 需要: $P(E|H), P(E|H^c), P(H)$
7. eg: A chopstick of unit length is broken at random into two pieces.
8. let R = shorter piece, find pdf of R .
9. let u = the point from 0, at which the chopstick is broken.
10. $R(u) = \min\{\frac{u}{1-u}, \frac{1-u}{u}\}$
11. The event $\{u: R(u) \leq r\}$ means $\{\frac{u}{1-u} \leq r\} \cup \{\frac{1-u}{u} \leq r\}$
12. $\Leftrightarrow \{0 < u \leq \frac{r}{1+r}\} \cup \{\frac{1}{1+r} \leq u < 1\}$
13. $P(R \leq r) = P(0 < u \leq \frac{r}{1+r}) + P(\frac{1}{1+r} \leq u < 1) = \frac{2r}{1+r}$
14. Derivative of $\frac{2r}{1+r}$ is $\frac{2}{(1+r)^2}$
15. $\therefore f(x) = \begin{cases} \frac{2}{(1+x)^2} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
16. 注意: cdf $\rightarrow$ pdf
17. joint pdf $f(x, y)$: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
18. $P(x \leq a, y \leq b) = \int_{-\infty}^a \int_{-\infty}^b f(x, y) dx dy$
19. $f(x, y) = \frac{\partial^2}{\partial x \partial y} P(X \leq x, Y \leq y)$
20. marginal pdf: $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$
21. $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$
22. $P(X \leq x) = \int_{-\infty}^x f_X(x) dx$
23. $P(Y \leq y) = \int_{-\infty}^y f_Y(y) dy$
24. X, Y are independent $\Leftrightarrow P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$
25. $f(x, y) = f_X(x)f_Y(y)$
26. wait for two buses the arrival times X, Y has a joint pdf: $f(x, y) = 6e^{-(2x+y)}$ for $x > 0$ and $y > 0$
27. (1) X, Y 独立? (2) 求 pdf of $\min\{X, Y\}$
28. $f_X(x) = 6e^{-2x} \int_0^{\infty} e^{-y} dy = 2e^{-2x}$
29. $f_Y(y) = 6e^{-y} \int_0^{\infty} e^{-2x} dx = 3e^{-y}$
30. (3) let $Z = \min\{X, Y\}$
31. $P(Z \leq t) = P(X \leq t \cup Y \leq t) = \int_0^t \int_0^{\infty} f(x, y) dx dy + \int_t^{\infty} \int_0^t f(x, y) dx dy$
32. $= \int_0^t \int_0^{\infty} 6e^{-(2x+y)} dx dy + \int_t^{\infty} \int_0^t 6e^{-(2x+y)} dx dy$
33. $= \int_0^t 3e^{-y} dy + \int_t^{\infty} 3e^{-y} dy = 1 - e^{-t}$
34. $\therefore f_Z(t) = e^{-t}$ for $t > 0$
35. $\therefore P(Z \leq t) = 1 - e^{-t}$
36. $f_Z(t) = e^{-t}$ for $t > 0$

Lec 5 Discrete RV

1. probability mass function (pmf): $P(x) = P(X=x)$
2. cumulative distribution function (cdf): $F(x) = P(X \leq x) = \sum_{x_i \leq x} P(x_i)$
3. joint probability mass function: $P(x, y) = P(X=x, Y=y)$
4. marginal pmfs: $P_X(x) = P(X=x) = \sum_y P(x, y) = \sum_y p(x, y)$
5. $P_Y(y) = P(Y=y) = \sum_x P(x, y) = \sum_x p(x, y)$
6. X, Y are independent $\Leftrightarrow P(X=x, Y=y) = P(X=x)P(Y=y)$
7. $P(x, y) = P_X(x)P_Y(y)$
8. Expectation
9. $E[X] = \sum_i x_i P(X=x_i) = \sum_i x_i p(x_i)$
10. $E[X+Y] = E[X] + E[Y]$ (in general)
11. $Z = X+Y$
12. $E[X+Y] = E[Z] = \sum_z z P(Z=z) = \sum_z z P(X+Y=z)$
13. $P(X+Y=z) = \sum_{x+y=z} P(x, y)$
14. $E[X_1 + X_2 + \dots + X_n] = E[X_1] + E[X_2] + \dots + E[X_n]$
15. eg: n 个孩子, 从第一个开始选, 选到第几个孩子 X = the number of children to be selected.
16. Let $I_k = \begin{cases} 1 & \text{if } k\text{-th child is taller than before} \\ 0 & \text{otherwise} \end{cases}$
17. $X = I_1 + I_2 + \dots + I_n$ $P(I_k=1) = \frac{1}{k}$
18. $E[X] = E[I_1] + E[I_2] + \dots + E[I_n] = \sum_{k=1}^n \frac{1}{k}$
19. 注意: ① 常用 $X_i = 0$ 或 1 来求解 X 求 $E[X]$
20. ② 常用于重复性动作出现的次数

3. 正态分布 $X \sim N(\mu, \sigma^2)$ Normal
 $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad x \in \mathbb{R}$
 $E(X) = \mu \quad \text{Var}(X) = \sigma^2 \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}}$ 是最大
4. 标准正态分布 $X \sim N(0, 1)$
 $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \quad x \in \mathbb{R}$
 $E(X) = \mu = 0 \quad \text{Var}(X) = \sigma^2 = 1$
 cdf: $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$ (查表)
- If $X \sim N(\mu, \sigma^2)$, then $aX+b \sim N(a\mu+b, a^2\sigma^2)$
 - If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$
 - If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent normal rv, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
 - μ 与 σ^2 对 $f(x)$ 图形的影响
 - $\sigma^2 \uparrow$ 图象越矮, 越分散
 - $\sigma^2 \downarrow$ 图象越高, 越集中
 - $\mu \uparrow$ 图象向右挪
 - $\mu \downarrow$ 图象向左挪

Problem & Methods

- eg. 证明类
 Show that if $E \subset F$, then $P(E) \leq P(F)$
 if $E \subset F$, then $F = E \cup (F \setminus E)$
 $P(F) = P(E) + P(F \setminus E) \geq P(E)$
- * 注意: 多拆解, 多用 "\", " $\subset$ " 读
- eg. 题型综合类: 条件 $\times$ 多个事件
 两个 station 播报天气 晴天或下雨
 已知 1 说晴, 2 说雨
 S : tomorrow is sunny
 R : tomorrow is rainy
 F_1 : station 1 predicts sunny } 顺着
 F_2 : station 2 predicts rainy } 题目
 条件
 $P(F_1|S) = 0.9 \quad P(F_2|S) = 0.2$
 $P(F_1|R) = 0.1 \quad P(F_2|R) = 0.8$
 $P(F_1 F_2|S) = P(F_1|S)P(F_2|S) \quad P(F_1 F_2|R) = P(F_1|R)P(F_2|R)$
 $P(S|F_1 F_2) = \frac{P(F_1 F_2|S)P(S)}{P(F_1 F_2|S)P(S) + P(F_1 F_2|R)P(R)}$

- eg. 分布性质在应用
 A, B, C 三个 machines. 2个同时工作才行.
 lifetime $(X_A, X_B, X_C) \sim \text{Exp}(\lambda)$
 先开 A 和 B. 用 C 替换不行的那一个.
 最后直到机器无法工作.
- (a) What is the expected time to stop?
 (b) What is the probability that at this time the only functional machine is C?
- (a) $T_1 = \min\{X_A, X_B\} \sim \text{Exp}(2\lambda)$ Property 2
 $T_2 = \min\{X_{\text{survivor}}, X_C\} \sim \text{Exp}(2\lambda)$ Property 1
- * 注意: 出现多个 $\text{Exp}(\lambda)$, 考虑 Property 2
 瓶颈时间, 考虑 Property 1 (无记忆)

$$E(T_1) = \frac{1}{2\lambda} \quad E(T_2) = \frac{1}{2\lambda}$$

$$T = T_1 + T_2 \quad E(T) = E(T_1) + E(T_2) = \frac{1}{\lambda}$$

(b) $P(X_{\text{survivor}} < X_C) = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{1}{2}$

求: $X \sim \text{Exp}(\lambda_1), Y \sim \text{Exp}(\lambda_2)$ 且相互独立
 $P(X < Y) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$
 $P(X < Y) = \int_0^\infty P(X < Y=y) f_Y(y) dy$ 全概率
 $= \int_0^\infty P(X < y) f_Y(y) dy$
 $= \int_0^\infty (1 - e^{-\lambda_1 y}) \lambda_2 e^{-\lambda_2 y} dy$
 $= \int_0^\infty \lambda_2 e^{-\lambda_2 y} - \int_0^\infty \lambda_2 e^{-(\lambda_1 + \lambda_2)y} dy$
 $= 1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{\lambda_2}{\lambda_1 + \lambda_2}$